Highlights

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- BERT models lead clinical NER; WER ≈9% achievable with Whisper/PyAnnote pipelines
- Communication markers (self-repair, engagement, barriers) predict therapeutic adherence
- Confirmed gap: clinical interactions rarely operationalised as speech-derived variables

Speech Analytics and Natural Language Processing Applied to Clinical Communication: A Scoping Review

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Abstract

Background: Professional-patient communication constitutes a dense source of clinical information that remains predominantly unstructured and analytically underutilised. Recent advances in Automatic Speech Recognition (ASR), Natural Language Processing (NLP) and vocal biomarker analysis have expanded the capacity to extract structured data from clinical interactions during anamnesis. Nevertheless, few studies operationalise these interactions as measurable variables derived from speech, particularly with regard to communicational and behavioural markers that can be used for data analysis and outcome assessment in patient care.

Objective: To identify, synthesise and critically evaluate the evidence on the application of *Speech Analytics* and NLP to professional-patient clinical communication, mapping methodological approaches, identified markers, technical challenges and gaps for future research.

Methods: Scoping review following the PRISMA-ScR guidelines (Tricco, Lillie, Zarin, O'Brien, Colquhoun, Levac, Moher, Peters, Horsley, Weeks, Hempel, Akl, Chang, McGowan, Stewart, Hartling, Aldcroft, Wilson, Garritty, Lewin, Godfrey, Macdonald, Langlois, Soares-Weiser, Moriarty, Clifford, Tunçalp and Straus, 2018) and JBI (Peters, Marnie, Tricco, Pollock, Munn, Alexander, McInerney, Godfrey and Khalil, 2020). The search was concentrated on the primary database PubMed/MEDLINE (filters: 2014-present, English/Spanish, free full text) and supplemented by manual *snowballing*. The final corpus comprises 24 articles selected from 786 records in the primary electronic search (after removing 27 duplicates) and 7 studies added manually by active search, organised into four thematic axes: (A) Clinical NLP and biomedical entity extraction; (B) automated clinical documentation and ASR; (C) physician-patient communication; (D) vocal biomarkers and multimodal analysis.

Results: Studies evidence consolidated advances in clinical NLP (*transformer*-based models, BERT leads Named Entity Recognition) and automated transcription (WER ≈9% achievable with Whisper/PyAnnote). Patient-centred communication demonstrates measurable impact on objective health outcomes. Communicational markers – self-repair, engagement, emotional support and barrier indicators – are identified as adherence predictors. Multimodal analysis (acoustic + semantic) remains in an early stage; most approaches focus on documentation automation, neglecting clinical interaction as an object of analysis in its own right.

Conclusions: A consistent gap is confirmed: few studies operationalise clinical interactions as measurable variables derived from speech, particularly communicational and behavioural markers extracted from the anamnesis. This review grounds the need for studies integrating *Speech Analytics* with structured analysis of clinical communication as a data source for adherence monitoring and quality of care.

1. Introduction

Clinical communication is one of the richest and least explored data sources in healthcare. In public health systems, anamnesis sessions and therapeutic guidance – verbal interactions with a propedeutic structure defined by clinical semiology (Porto and Porto, 2019) – occur predominantly as analytically unstructured exchanges, rarely converted into measurable data (Wang et al., 2018; Koleck et al., 2019).

The quality of the professional-patient interaction has a measurable impact on adherence and objective clinical outcomes (Kelley et al., 2014). Communication barriers – including health literacy gaps, misaligned expectations and miscommunication patterns – are associated with low therapeutic adherence and worse prognosis (McCabe and Healey, 2018; Alkhamees and Alasqah, 2023).

Speech Analytics – the computational analysis of spoken language – offers a pathway to extract structured and measurable indicators from these interactions. Recent advances in ASR, particularly transformer-based models such as Whisper (Radford et al., 2023), combined with instruction-tuned large language models (LLMs), have lowered the barriers for deploying end-to-end NLP *pipelines* in healthcare (Klusty et al., 2025). Nevertheless, most existing approaches target diagnostic support or documentation automation, without treating the clinical interaction itself as an object of structured analysis (van Buchem et al., 2021). The development of a *Speech Analytics* protocol for monitoring orthopedic rehabilitation in a public health system (Baena Neto and Becerra Sablón, 2026) exemplifies the translational potential of these technologies and underscores the need for a consolidated evidence map to support the design of prospective studies.

Recent reviews have addressed aspects partially overlapping with this scope. Falcetta, de Almeida, Lemos, Goldim and da Costa (2023) identified the absence of prospective validation in automated documentation systems for clinical interactions. Ng, Wang, Zhou, Zhou, Goh, Sim, Tan, Goh and Ng (2025) evaluated the technical performance of ASR systems for clinical documentation, reporting WER values between 8.7% and over 50% depending on the context. Alabbad and Alhoshan (2026) mapped ASR applications in healthcare in the post-LLM era. Albert et al. (2024) conducted a scoping review on AI, speech and NLP methods for clinician-patient communication assessment. None of these reviews, however, focuses on the operationalisation of the clinical interaction as a set of **speech-derived measurable variables** – particularly communicational and behavioural markers (self-repair, engagement, barriers, health literacy) – as an object of analysis in its own right, independent of documentation or diagnosis. It is precisely this gap that the present review aims to map.

This scoping review aims to map the evidence on *Speech Analytics* and NLP applied to clinical communication, characterising the methodological landscape and identifying gaps for future research.

2. Methods

2.1. Study design

This scoping review follows the methodological framework proposed by Arksey and O'Malley (2005), refined by Levac, Colquhoun and O'Brien (2010) and the Joanna Briggs Institute (JBI) (Peters et al., 2020). Conduct and reporting follow the PRISMA-ScR guidelines (*Preferred Reporting Items for Systematic Reviews and Meta-Analyses* extension for Scoping Reviews) (Tricco et al., 2018).

2.2. Research questions

- **Q1:** Which *Speech Analytics* approaches have been applied to clinical communication over the past five years?

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- **Q2:** Which NLP and ASR methods are used for automated transcription and analysis of clinical consultations?
- **Q3:** Which communicational and behavioural markers have been identified or proposed in the professional-patient interaction?
- **Q4:** What are the technical, ethical and methodological challenges for automated analysis of clinical communication?

2.3. Eligibility criteria

2.3.1. Inclusion criteria

- **IC1:** The study applies *Speech Analytics*, NLP or ASR technologies for clinical transcription and extraction of unstructured medical data, or analyses and measures the foundations of professional-patient communication.
- **IC2:** The study focuses on free-text EHR interactions, spoken clinical dialogues (consultations), automated documentation generation (digital scribes/SOAP notes) or theoretical frameworks of shared decision-making.
- **IC3:** Full-text availability in English, Portuguese or Spanish; published between January 2014 and 2026.
- **IC4:** Presents original development, empirical feasibility testing, algorithm validation, or constitutes a prior systematic or scoping review on the topic.

2.3.2. Exclusion criteria

- **EC1:** Studies that do not address NLP, ASR or audio analysis of clinical interactions.
- **EC2:** Studies describing transcription or extraction systems without analysing interaction quality or impact on the professional-patient relationship.
- **EC3:** Exclusive focus on voice-based diagnosis of physical or psychiatric conditions without analysis of dialogic interaction.
- **EC4:** Opinion pieces, commentaries, editorials, conference abstracts or books.
- **EC5:** Full text unavailable without payment or inaccessible through institutional channels.
- **EC6:** Studies that do not describe data collection, processing methodology or technical validation.
- **EC7:** Studies addressing exclusively non-clinical interactions or simulated contexts without demonstrated practical transferability.

2.4. Information sources and search strategy

The literature search was conducted independently by three different researchers, each using multiple search strategies to cover the thematic axes of the review. The primary database used was PubMed/MEDLINE. The following search filters were applied: publication period from January 2014 to the present (2026), articles published in English and Spanish, and availability of free full text.

Each researcher used more than one search string. The specific strings used were as follows:

1. **String 1 (NLP, Symptoms and EHR):**

```
("natural language processing"[Title/Abstract])
AND
(symptoms[Title/Abstract] OR "patient-reported symptoms"[Title/Abstract])
AND
("electronic health records"[Title/Abstract] OR "patient-centered"[Title/Abstract])
```

2. **String 2 (Clinical Information Extraction and NER):**

```
("clinical information extraction"[Title/Abstract]
OR "medical information extraction"[Title/Abstract]
OR "named entity recognition"[Title/Abstract])
AND
("medical free text"[Title/Abstract]
OR "review"[Title/Abstract]
OR "patient-generated health data"[Title/Abstract])
```

3. **String 3 (Clinician-Patient Communication and Outcomes):**

```
("patient-physician communication"[Title/Abstract]
OR "doctor-patient communication"[Title/Abstract]
OR "patient-clinician relationship"[Title/Abstract])
AND
("intercultural settings"[Title/Abstract]
OR miscommunication[Title/Abstract]
OR "healthcare outcomes"[Title/Abstract]
OR "shared decision-making"[Title/Abstract]
OR "communication quality scale"[Title/Abstract])
```

4. **String 4 (Vocal Biomarkers and Speech Analytics):**

```
("vocal biomarkers"[Title/Abstract] OR "speech analytics"[Title/Abstract])
AND
("clinical practice"[Title/Abstract]
OR "voice for health"[Title/Abstract]
OR "SUS"[Title/Abstract])
```

5. **String 5 (NLP, Speech and Reviews):**

```
(("Applying natural language processing"[Title/Abstract])
  AND ("Systematic review"[Title/Abstract]))
OR
(("Identifying Topics"[Title/Abstract])
  AND ("Natural Language Processing Methods"[Title/Abstract]))
OR
(("Automatic speech recognition"[Title/Abstract])
  AND ("patient-clinician conversations"[Title/Abstract]))
```

Additionally, a comprehensive search was conducted using the following unified general string:

```
("speech analytics" OR "speech analysis" OR "speech recognition"
OR "automatic speech recognition" OR ASR OR "digital scribe"
OR "clinical transcription" OR "clinical documentation" OR "vocal biomarkers"
OR "healthcare conversations" OR "doctor-patient conversations"
OR "patient-doctor conversations")
AND
("natural language processing" OR NLP OR "large language model" OR LLM
OR "information extraction" OR "clinical information extraction"
OR "named entity recognition" OR NER OR "relation extraction" OR "text mining"
OR "conversation analysis" OR "topic modeling" OR "SOAP note" OR "SOAP notes")
```

```
AND
(healthcare OR clinical OR medicine OR medical OR physician OR clinician
OR doctor OR patient OR "electronic health record" OR EHR OR hospital
OR outpatient)
```

To mitigate the loss of relevant articles and qualitatively enrich the corpus, the electronic search was supplemented by the *snowballing* method (manual reference tracking or *backward snowballing*). From the systematic reviews and foundational articles located in the primary search, their reference lists were manually analysed to identify and include other studies that strictly meet the defined eligibility criteria, ensuring coverage of key works of methodological or conceptual relevance to this review.

2.5. Study selection

Records retrieved from the primary PubMed/MEDLINE search were imported into Rayyan (Ouzzani, Hammady, Fedorowicz and Elmagarmid, 2016) for deduplication and screening. Titles and abstracts were screened independently by two reviewers (F.E.S. and M.M.), with agreement assessed using Cohen's kappa coefficient ($\kappa = 0.78$). Disagreements were resolved by a third reviewer (L.K.T.I.) or by the corresponding author. Full texts of potentially eligible records were then retrieved and assessed against the eligibility criteria. The selection process is documented in the PRISMA-ScR flowchart (Fig. 1).

2.6. Data extraction

Data were extracted using a standardised form containing: author(s), year, country, study design, clinical context, population characteristics, technology used (ASR model, NLP framework, LLM), extracted clinical variables, reported outcomes, ethical framework and limitations.

2.7. Synthesis

A narrative synthesis was performed. Findings are organised around the four thematic axes defined in the eligibility criteria: (A) Clinical NLP and biomedical entity extraction; (B) automated clinical documentation and ASR; (C) physician-patient communication; and (D) vocal biomarkers and multimodal analysis.

3. Results

3.1. Study selection

Searches in the primary database (PubMed/MEDLINE) retrieved 786 records. After removing 27 duplicates, 759 records were screened by title and abstract, resulting in 35 articles selected for full-text assessment. Of these, 17 were excluded for not meeting the eligibility criteria: 8 for not addressing NLP, ASR or audio analysis of clinical interactions (EC1), 5 for describing transcription or extraction systems without analysing the quality of the professional-patient interaction (EC2), and 4 for consisting of opinion pieces, editorials or normative documents without empirical data (EC4). The 7 articles identified by *snowballing* (backward reference tracking of included systematic reviews) were subjected to the same eligibility assessment independently of the electronic records; all met the inclusion criteria and none was excluded. In total, 24 articles were included in the synthesis (17 from the electronic search + 7 from *snowballing*). The PRISMA-ScR flowchart is presented in Fig. 1.

3.2. Characteristics of included studies

The 24 studies in the corpus span the period 2014–2025 (Table 1).

Table 1: Characteristics of studies included in the synthesis.

Code	Author, year	Type	Axis	Technology	Main outcome	Country
S01	Bazoge et al., 2023	Syst. rev.	A	Transformers, BERT	NER in CDWs; phenotyping	France

(continued)

Code	Author, year	Type	Axis	Technology	Main outcome	Country
S02	Navarro et al., 2023	Syst. rev.	A	BERT, BioBERT	Clinical NER (problems, treatments)	Australia
S03	Watabe et al., 2025	Empirical	A	BERT-CRF	F1 > 0.8; reported symptoms	Japan
S04	Sezgin et al., 2023	Empirical	A	NER + SNOMED CT (zero-shot)	Symptom identification in PGHD	USA
S05	Koleck et al., 2019	Syst. rev.	A	NLP, classifiers	Symptom extraction from EHR	USA
S06	Wang et al., 2018	Lit. rev.	A	cTAKES, MedLEE, ML	263 clinical extraction systems	USA
S07	Krishna et al., 2021	Preprint‡	B	CLUSTER2SENT	SOAP notes generated from consultations	USA
S08	Klusty et al., 2025	Empirical	B	Whisper + PyAnnote	On-premise transcription; privacy	USA
S09	Abbasian et al., 2024	Framework	B	Generative AI, RAG	Groundedness, safety, empathy	USA
S10	Aleksić et al., 2017	Empirical	B	EHR summarisation	Dashboard for chronic disease	Serbia
S11	Tran et al., 2023	Empirical	B	General vs. specialist ASR	WER; errors in short pronouns	USA
S12	Schloss & Konam, 2020	Preprint‡	B	Hierarchical Bi-LSTM	Utterance classification → SOAP	USA
S13	van Buchem et al., 2021	Scoping rev.	B	Digital scribes (review)	Usability gaps in real clinical use	Netherlands
S14	Falcetta et al., 2023	Syst. rev.	B	Scribes (review)	Mapping of 22 scribes; no prospective validation	Brazil
S15	Ng et al., 2025	Syst. rev.	B	Clinical ASR (review)	WER ranged from 8.7% to >50% by context	Singapore
S16	Shao et al., 2025	Empirical	C	DPCQ scale (14 items)	Physician-patient communication quality	China

(continued)

Code	Author, year	Type	Axis	Technology	Main outcome	Country
S17	Muscat et al., 2020	Framework	C	Shared decision-making	Health literacy; engagement	Australia
S18	Kelley et al., 2014	Meta-analysis	C	–	Clinical relationship → objective outcomes	USA
S19	Rucinski et al., 2022	Syst. rev.	C	Adherence metrics	Bilateral adherence in orthopedics	USA
S20	McCabe & Healey, 2018	Empirical	C	Conversation analysis	Self-repair as adherence predictor	UK
S21	Alkhamees & Alasqah, 2023	Integr. rev.	C	–	Intercultural barriers in communication	Saudi A.
S22	Fagherazzi et al., 2021	Framework	D	Vocal pipeline (acoustics+NLP)	Vocal biomarkers of health and emotion	Luxembourg
S23	Albert et al., 2024	Scoping rev.‡	D	AI, Speech, NLP	Clinician-patient communication via AI	France
S24	Imel et al., 2022	Report	D	Sequential neural networks	Emotional valence; human rater agreement	USA

‡Preprint/not peer-reviewed (arXiv/medRxiv) — included as they meet IC1–IC4; preprint acceptance per Elsevier/IJMI policy.

3.3. Axis A —Clinical NLP and biomedical entity extraction

The studies in Axis A map the evolution of NLP approaches applied to clinical information extraction, documenting the transition from rule-based systems to deep learning architectures and *transformer* models.

The historical landscape of the field is described by Wang et al. (2018), whose review of 263 studies evidences the predominance of symbolic systems such as cTAKES and MedLEE in the processing of electronic health records. The authors demonstrate how adherence to linguistically rule-based approaches was the norm until the arrival of machine learning techniques. The consolidation of the transition to deep learning models is documented by Bazoge, Morin, Daille and Gourraud (2023), in a systematic review on the use of NLP in clinical data warehouses (CDWs): the authors evidence the migration from symbolic methods to *transformers*, identifying their applicability in phenotyping and symptom management tasks from secondary data.

In the specific domain of biomedical Named Entity Recognition (NER), Navarro et al. (2023) systematically reviewed the literature on NER and relation extraction in free-text medical records and concluded that models such as BERT lead in extracting the categories of "problems" and "treatments". The authors point out, however, the scarcity of implementations in real clinical settings as a persistent limitation of the field.

The ability of *transformer* models to capture patient colloquial language was demonstrated by Watabe et al. (2025), who developed and validated a BERT-CRF model for extracting symptoms reported in everyday language, achieving an F1-score above 0.8. The study highlights the need for *fine-tuning* with authentic narrative data to capture slang and linguistic nuances present in patient speech. Complementarily, Sezgin et al. (2023) demonstrated the feasibility of a *zero-shot* approach combining NER and the SNOMED CT ontology to identify symptoms in patient-generated health data (PGHD), without specific retraining, validating the possibility of pre-trained models translating lay language into structured clinical metrics.

At the intersection of NLP and electronic health records, Koleck et al. (2019) synthesised the efficacy of classification algorithms for extracting subjective symptoms – such as pain and mood changes – from free-text EHR narratives, evidencing the technical feasibility of automated extraction even for subjective dimensions.

Taken together, the studies in Axis A confirm the technical maturity of clinical NLP for biomedical entity extraction, with BERT and its variants established as the state of the art. The cross-cutting gap identified – highlighted by Navarro et al. (2023) – is the distance between performance in laboratory settings and deployment in real clinical systems, where data privacy, linguistic variability and integration with care workflows remain unsolved challenges.

3.4. Axis B —Automated clinical documentation and ASR

The studies in Axis B cover the full cycle of automated clinical documentation: from speech recognition to the generation of structured records, including the technical challenges and ethical requirements for hospital deployment.

The foundations of automated documentation are established by Aleksić et al. (2017), whose electronic health record summarisation model demonstrates the feasibility of dashboards capable of identifying patients requiring specialist attention, validating readable synthesis as a strategy to reduce clinician cognitive load.

In the specific domain of SOAP note generation, two studies address complementary strategies. Schloss and Konam (2020) used hierarchical Bi-LSTM networks to classify consultation utterances into SOAP sections with high accuracy, proposing modular strategies to handle data "noise" in automatic transcriptions. Krishna et al. (2021), in turn, proposed the CLUSTER2SENT model, based on modular summarisation, to generate structured notes from dense physician-patient conversations, facilitating the retrieval of verbal evidence by the clinician in the analytical output.

With regard to automatic transcription, Klusty et al. (2025) described a system based on OpenAI Whisper and PyAnnote operating on closed (*on-premise*) servers, simultaneously resolving the speaker diarisation challenge and the privacy requirements for hospital deployment. Complementarily, Tran et al. (2023) demonstrated that errors in short pronouns (such as "I" and "you") are more critical for the validity of clinical indicators than errors in complex medical terms, highlighting the need for accuracy in casual language to ensure the quality of extracted data.

Generative AI-based approaches bring a distinct set of metric requirements. Abbasian et al. (2024) propose safety, factual validity (*groundedness*) and empathy metrics as mandatory indicators for evaluating health conversations generated by language models, grounding the introduction of Retrieval-Augmented Generation (RAG) as a mechanism to anchor model responses in facts extracted from the consultation and mitigate algorithmic hallucinations.

The state of the field is synthesised by van Buchem et al. (2021), in a scoping review on the deployment of digital scribes in clinical practice: the authors point to the absence of prospective usability and real clinical utility studies as the central gap in the field, with most initiatives still limited to proof-of-concept without prospective clinical validation.

Taken together, the studies in Axis B reveal a field in rapid technical evolution (with ASR, diarisation and SOAP note generation *pipelines* progressively more robust), but with prospective clinical validation still scarce (van Buchem et al., 2021). Most approaches direct processing towards documentation automation, subordinating the analysis of clinical interaction quality to the production of the structured record.

3.5. Axis C —Physician-patient communication

The studies in Axis C establish the empirical basis for professional-patient communication quality as a determinant of clinical outcomes, identifying the measurable dimensions of the therapeutic interaction and the barriers that compromise its effectiveness.

The impact of the clinician-patient relationship on objective health outcomes is demonstrated by the meta-analysis of Kelley et al. (2014), which confirms the significant effect of the emotional and cognitive dimensions of communication on measurable clinical results, providing the theoretical foundation for investing in automatic quality metrics of the therapeutic relationship.

In the domain of structured measurement, Shao et al. (2025) developed and validated the 14-item DPCQ (*Doctor-Patient Communication Quality*) scale, focused on emotional support and health education. The semantic categories proposed by the authors (emotional interaction versus behavioural guidance) are directly applicable as targets for speech-derived automatic metrics.

The dimension of therapeutic adherence in the orthopedic context is addressed by Rucinski et al. (2022), whose systematic review demonstrates that current metrics fail by ignoring the barriers imposed by the healthcare team and the system, not only by the patient. The authors argue for the creation of auditable data capable of mapping bilateral failures in orthopedic rehabilitation, an argument that reinforces the need for instruments to analyse the interaction, not just the isolated behaviour of the patient.

At the microscopic level of speech, McCabe and Healey (2018) demonstrated that the use of "self-repair" by physicians (i.e., the spontaneous correction of utterances during the consultation) predicts superior treatment adherence. The finding validates the analysis of low-level linguistic dynamics as predictors of clinical outcomes, demonstrating that automatically identifiable communicational markers can carry prognostic value.

The role of health literacy in effective communication is analysed by Muscat et al. (2020), who demonstrate that shared decision-making requires the active building of patient skills through effective verbal instruction, in contrast with the passive informational model. Speech monitoring in the clinical context must therefore capture indicators of communicational literacy success, not merely unilateral information transfer.

Linguistic and cultural diversity contexts introduce an additional layer of complexity. Alkhamees and Alasqah (2023) mapped how linguistic disparities and authoritarian consultation styles generate patient inhibition and compromise the quality of clinical decisions. In the diverse context of SUS (Brazil's National Health System), these barriers often remain invisible to managers, but are potentially identifiable through automated interaction analysis.

The clinical propedeutics systematised by Porto and Porto (2019) organises the anamnesis as a structured sequence of inquiry elements, defining the propedeutic characteristics of the complaint – location, intensity, quality, radiation, triggering and relieving factors, duration and evolution – as mandatory semantic targets of diagnostic reasoning. This structure constitutes the clinical grammar of verbal interaction: the set of information that the clinician must elicit and the patient must verbalise for the anamnesis to fulfil its diagnostic function. None of the studies in this review's corpus crosses the propedeutic model with automatic analysis of verbal interaction: the studies in Axis C measure communication quality in its relational and cognitive dimensions, but do not verify whether the propedeutic characteristics were effectively verbalised, understood and recorded during the consultation.

Taken together, the studies in Axis C provide the map of what must be measured in clinical interaction: emotional and cognitive dimensions (Kelley et al., 2014; Shao et al., 2025), linguistic adherence markers (McCabe and Healey, 2018; Rucinski et al., 2022), health literacy competencies (Muscat et al., 2020) and intercultural barriers (Alkhamees and Alasqah, 2023). The emerging gap is twofold: no study operationalises these dimensions as automatically speech-derived variables (technological gap), and no study verifies whether the propedeutic characteristics of the anamnesis (Porto and Porto, 2019) were elicited with adequate communicational quality during real interaction (clinical-methodological gap). Measurement remains dependent on self-report scales or direct observation, without integration with acoustic-semantic analysis *pipelines* guided by the clinical structure of the anamnesis.

3.6. Axis D —Vocal biomarkers and multimodal analysis

The studies in Axis D explore the acoustic and prosodic dimension of clinical communication, extending the analysis beyond the semantic content of the transcribed text to the vocal signals that accompany speech.

Fagherazzi et al. (2021) describe the *pipeline* for capturing linguistic and acoustic features – prosody, vocal quality and rhythm – that indicate health states and emotional states of the speaker. The authors demonstrate that signals derived directly from the voice can function as clinical biomarkers, grounding the idea that the speech of both patient and clinician, beyond its semantic content, carries usable analytical information independently of textual transcription.

In the domain of real clinical consultation analysis, Imel et al. (2022) mapped emotional valences from textual transcripts of physician-patient dialogues using sequential neural networks, achieving agreement with trained human raters. Although the study demonstrates that emotional valence markers of verbal interaction can be automatically identified at utterance level with reliability comparable to human assessment (correlation of 0.60), the authors note that the model was trained and evaluated exclusively on the textual content of the dialogues (transcription diaries), without directly processing the audio signal or the acoustic layer.

The unresolved frontier of Axis D is identified by Albert et al. (2024), in a scoping review on AI, speech and NLP methods for clinician-patient communication assessment: the authors highlight the scarcity of studies on non-verbal language, voice tones and silences. Most existing approaches focus on transcription and semantic analysis, neglecting the acoustic-prosodic layer as an independent object of analysis. Silences, pauses and voice modulations – therapeutic and relational components of the anamnesis – remain without a consolidated method for automatic extraction and interpretation.

Taken together, the studies in Axis D reveal a field in an early stage of development: the technical foundations of voice processing exist (Fagherazzi et al., 2021) and there is feasibility in mapping emotional markers based on textual transcripts of real interactions (Imel et al., 2022), but the multimodal integration that truly unites the acoustic aspects of voice with the semantic aspects of text in clinical communication remains incipient (Albert et al., 2024). This gap acquires precise clinical contour when confronted with the propedeutics of Porto and Porto (2019): if the propedeutic characteristics of the complaint have acoustic-prosodic correlates in speech (vocal tension for pain, hesitation and pauses for doubt, prosodic disengagement for barriers), no study in the corpus guides the extraction of vocal biomarkers by the clinical grammar of the anamnesis. The integration between the propedeutic model and the acoustic-semantic *pipeline* represents the most unexplored methodological frontier of this review.

4. Discussion

Two parallel axes without systematic convergence

The mapped studies reveal two lines of development advancing in parallel without systematic convergence in the literature. The first line, formed by clinical NLP studies (Axis A) and automated documentation studies (Axis B), treats speech as input for documentation: the clinical encounter is transcribed to generate SOAP notes, identify biomedical entities or populate electronic health record fields (Bazoge et al., 2023; van Buchem et al., 2021; Falcetta et al., 2023; Ng et al., 2025). The second line, formed by physician-patient communication studies (Axis C) and vocal biomarker studies (Axis D), establishes what communication means clinically and demonstrates that acoustic signals carry diagnostic information (Kelley et al., 2014; Fagherazzi et al., 2021; Albert et al., 2024). In this tradition, communication quality is operationalised by self-report instruments or in-person observation, and vocal biomarkers are applied predominantly to the detection of health states (Parkinson's disease, depression, COVID-19), not to the communicational dynamics during the consultation.

The methodological heterogeneity reflects this structural divide. NLP studies such as Bazoge et al. (2023) and Wang et al. (2018) employ entity extraction pipelines on retrospective EHR text; ASR studies such as van Buchem et al. (2021) and Klusty et al. (2025) measure transcription accuracy and documentation time reduction; communication studies such as Shao et al. (2025) and Kelley et al. (2014) derive relational indicators through validated questionnaires. These traditions share the locus of clinical interaction but do not share measurement objects, making quantitative synthesis unfeasible with the studies in the current corpus.

Confirmed gap: anamnesis as the object of analysis

No study in the corpus operationalises the anamnesis as a structured clinical event for the automatic extraction of speech-derived communicational and behavioural markers. The distinction is critical: most studies treat the clinical interaction as a *means* – to produce documentation or detect pathologies – not as an *object* of communicational analysis. Krishna et al. (2021) and Schloss and Konam (2020) classify utterances by SOAP structure, not by the relational quality of the encounter. Abbasian et al. (2024) evaluate the efficacy of conversations generated by generative AI, not the communicative behaviour of the healthcare professional in real context. Shao et al. (2025) and McCabe and Healey (2018) describe relevant dimensions of communicative quality – clarity, empathy, comprehension barriers – but operationalise them through self-report instruments, not through automatically extracted speech markers.

The review by Albert et al. (2024) is the one that comes closest to this intersection: it maps SA and NLP methods for clinical communication assessment, but confirms the scarcity of studies linking acoustic-prosodic markers to measurable communicational outcomes, particularly in rehabilitation contexts. Porto and Porto (2019) offers a precise semiological framework for orthopedic anamnesis, including symptom hierarchy, chronology and functional exploration, which could guide the operational definition of these markers; however, no study retrieved in this review integrates that framework with automated speech analysis.

Implications for prospective studies

The identified gap points to a set of requirements for future studies: (i) recording of real clinical encounters under ecological conditions; (ii) ASR with models fine-tuned to Brazilian Portuguese; (iii) extraction of communicational markers (expression of doubt, pain, engagement, linguistic barriers) anchored in the semiological framework of the anamnesis (Porto and Porto, 2019); and (iv) correlation of these markers with adherence outcomes and quality of care in longitudinal follow-up. Baena Neto and Becerra Sablón (2026) propose exactly this integration: a prospective observational study in orthopedic rehabilitation consultations at SUS (Brazil's National Health System), with transcription via ASR (Whisper (Radford et al., 2023)) in on-premise processing and marker extraction for longitudinal outcome analysis. This approach is anchored in the technical repertoires of Axes A, B and D and in the communicational constructs of Axis C, integrating them in an unprecedented manner around the anamnesis as the primary unit of analysis.

Limitations of this review

This review presents four central limitations. First, although the electronic search was concentrated in PubMed/MEDLINE as the primary database, the inclusion of additional articles via *snowballing* (including relevant systematic reviews and reference studies) aimed to mitigate the loss of essential articles; however, the absence of automated searches in other indexed electronic databases (such as Scopus and Web of Science) remains a limitation for full generalisation of the literature. Second, there is a language bias: the mapped literature is predominantly in English, with under-representation of studies conducted in the Brazilian health context and in Portuguese, which may underestimate local Speech Analytics initiatives applied to SUS. Third, the selection of search filters in the primary database, such as the requirement of free full text, may have excluded relevant studies protected by paywall or journal subscription restrictions. Fourth, the heterogeneity of study type – including systematic reviews, empirical studies, preprints and conceptual models – prevents the conduct of a quantitative synthesis or structured meta-analysis, also limiting direct comparisons of methodological quality between approaches.

Ethical considerations

The application of Speech Analytics in clinical consultations raises ethical questions that go beyond standard human research considerations. Voice constitutes sensitive biometric data under the terms of the General Personal Data Protection Law (LGPD, Law No. 13.709/2018), imposing obligations of data minimisation, pseudonymisation, separation between identified and analytical datasets, and restricted access control. Research must be conducted in compliance with CNS Resolution No. 466/2012 (Conselho Nacional de Saúde, 2012) and CNS Resolution No. 510/2016 (Conselho Nacional de Saúde, 2016), and with Law No. 14.874/2024 (Brasil, 2024), which updates the regulatory framework for clinical research in Brazil. The Free and Informed Consent Form (TCLE – *Termo de Consentimento Livre e Esclarecido*) is mandatory for voice recording and cannot be waived when speech itself constitutes the object of analysis.

From a technical standpoint, Klusty et al. (2025) demonstrate that on-premise processing is feasible in a hospital environment and represents the most appropriate solution for privacy, eliminating the transmission of sensitive data to external servers. This approach aligns with the five principles for responsible AI proposed by Floridi, Cowls, Beltrametti, Chatila, Chazerand, Dignum, Luetge, Madelin, Pagallo, Rossi, Schafer, Valcke and Vayena (2018) (beneficence, non-maleficence, autonomy, justice and explicability), and with the international recommendations of the OECD (OECD, 2024) and the World Health Organization (World

Health Organization, 2024) for ethical governance of large-scale AI models in healthcare. The research protocol must also explicitly prohibit the use of extracted markers for autonomous clinical decision-making, preserving the decision-making role of the healthcare professional and the informed autonomy of the patient.

5. Conclusion

This scoping review mapped the state of the art in Speech Analytics and clinical communication with the aim of establishing the scientific foundation for prospective studies that integrate automated speech analysis with the assessment of communicational quality in orthopedic rehabilitation contexts.

The four thematic axes evidence consolidated advances in clinical NLP and ASR: biomedical entity extraction *pipelines* achieve competitive performance in electronic health records (Bazoge et al., 2023; Koleck et al., 2019), speech recognition models such as Whisper (Radford et al., 2023) enable the transcription of clinical encounters in hospital settings (Klusty et al., 2025; Ng et al., 2025), and vocal biomarkers demonstrate the capacity to capture physiological and emotional states relevant to healthcare (Fagherazzi et al., 2021; Albert et al., 2024). The physician-patient communication literature, in turn, consistently establishes that the relational quality of the clinical encounter influences therapeutic adherence and health outcomes (Kelley et al., 2014; Rucinski et al., 2022).

However, the review confirms a consistent and specific gap: few studies operationalise clinical interactions as a source of speech-derived measurable variables, particularly communicational and behavioural markers (expression of doubt, pain, engagement, comprehension barriers) automatically extracted during the anamnesis as a structured event. Most initiatives treat speech as input for documentation or as a signal for pathology detection, without anchoring the analysis in the semiological framework of the anamnesis (Porto and Porto, 2019), which defines what, in speech, is clinically significant from a communicational standpoint.

This gap points to the need for prospective studies that integrate Speech Analytics with structured analysis of clinical communication for real-time adherence monitoring and quality of care. Such studies require, in addition to adequate technical infrastructure, a robust ethical protocol that treats voice as sensitive biometric data, ensures informed consent and preserves patient privacy through local data processing. The methodological protocol described in Baena Neto and Becerra Sablón (2026) represents a direct response to this set of requirements in the context of SUS (Brazil's National Health System).

Declarations

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Conflict of interest

None of the authors has employment, consultancy, equity, honoraria, patents or funding ties to entities with commercial or institutional interest in the topic of this review. The prospective study protocol cited in the text (Baena Neto and Becerra Sablón, 2026) is co-authored by the corresponding author (R.B.N.) and the senior co-author (V.I.B.S.); this relationship is fully disclosed in the author list and does not constitute a conflict of interest under ICMJE guidelines.

Ethical approval

Not applicable. This study consists of a scoping review based exclusively on published and publicly accessible scientific literature. No primary data were collected, no participants were recruited, and no intervention was performed on human subjects. Under the terms of CNS Resolution No. 510/2016 (Conselho Nacional de Saúde, 2016), studies using public domain information are exempt from registration and assessment by the CEP/CONEP system.

Data availability

The data supporting the findings of this review are available as follows: (i) the **complete search strings** per database are registered in supplementary file S1 and in the project's public repository (https://doi.org/10.5281/zenodo.20821063); (ii) the **standardised data extraction form** (fields extracted per study, including authors, year, study design, population, technology used, outcomes and methodological quality assessment) is available as supplementary material S2; (iii) **screening decisions** (inclusion/exclusion by title, abstract and full text), with the respective reasons for exclusion, were recorded in the Rayyan platform (Ouzzani et al., 2016) and are made available upon request to the corresponding author; (iv) the **complete list of included studies**, with classification by thematic axis and quality score, is incorporated in Table 1 in the body of the article. Raw data from recordings, transcriptions or patient records do not apply to this study, which is based exclusively on published literature.

Use of artificial intelligence

In accordance with Elsevier's Generative AI Policy, the authors declare that the following artificial intelligence models were used in the preparation of this work: (i) **Claude Sonnet 4.6** (Anthropic PBC, https://claude.ai, 2025–2026), used as an assistant in literature searches in electronic databases; (ii) **Claude Code** (Anthropic PBC, https://claude.ai/code, 2026), used to support corpus screening, data extraction and translation of excerpts from articles consulted in a foreign language. All intellectual content, analyses, syntheses and conclusions are the sole responsibility of the authors, who fully reviewed and validated all material described that was produced with AI assistance.

CRediT authorship contribution statement

Rafael Baena Neto: Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft. **Vicente Idalberto Becerra Sablón:** Supervision, Conceptualization, Writing – review & editing. **Fabrício Evangelista dos Santos:** Investigation, Data curation, Methodology, Software, Writing – review & editing. **Lucas Kenzo Terazzin Ida:** Investigation, Data curation, Formal analysis, Writing – review & editing. **Murilo Nogueira Minutti:** Investigation, Data curation, Formal analysis, Visualization, Writing – review & editing.

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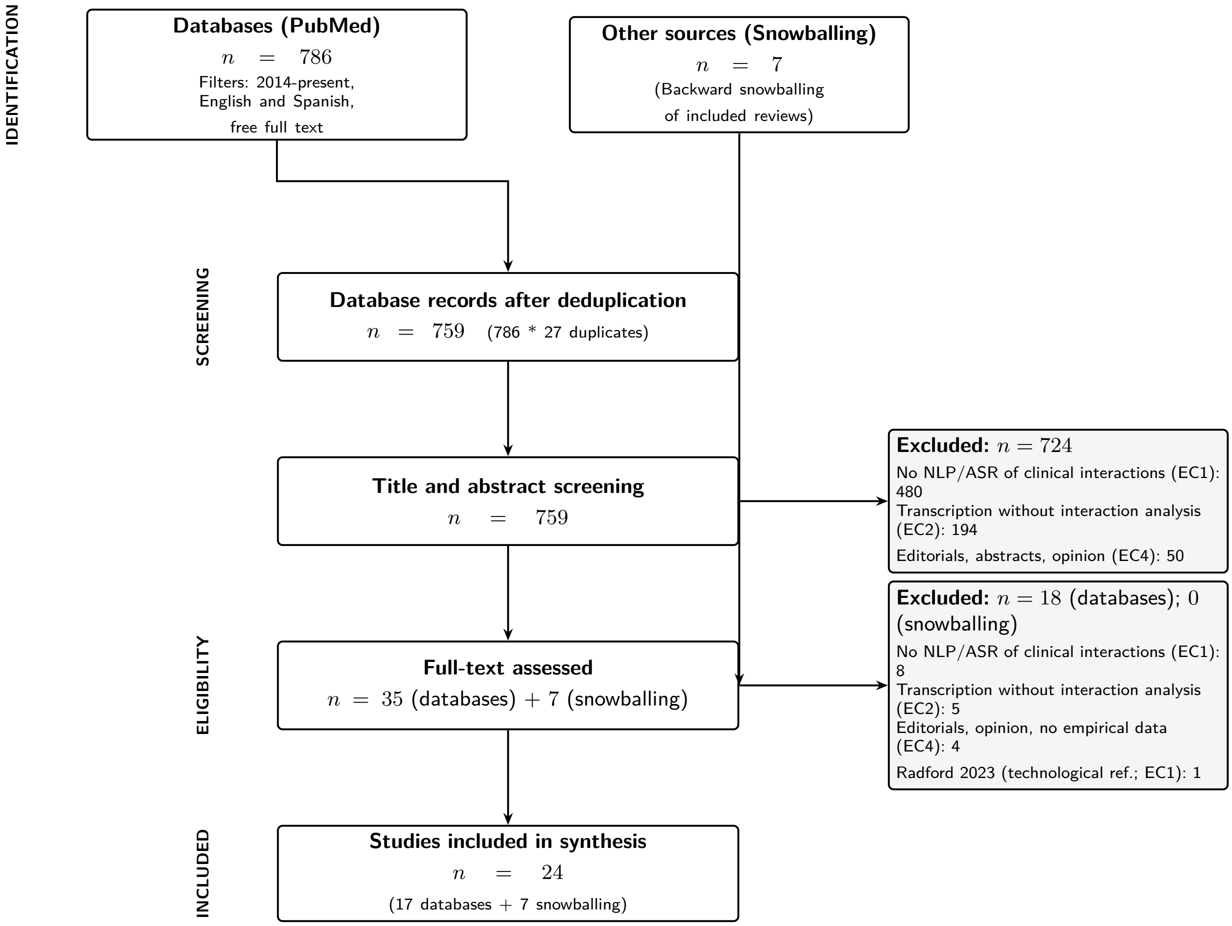

Exact counts: PubMed/MEDLINE, period 2014-present, free full text filter. Snowballing assessed with the same eligibility criteria. $\kappa = 0.78$ (F.E.S. and M.M.; Cohen's kappa).

Figure 1: PRISMA-ScR flowchart of the study selection process.